

Homework 3

2.2 Draw the DC and AC circuits of each amplifier circuit as shown in figure P2.2. Assume all capacitors have no resistance to AC signals.

2.3 For circuits (a)(b) in fig P2.2, What kind of amplifier are they? (common emitter, common collector, Common base), Write the expression of Q , A_u , R_i , R_o

2.2 画出图 P2.2 所示各电路的直流通路和交流通路。设图中所有电容对交流信号均可视为短路。

2.3 分别判断图 P2.2 (a)、(b) 所示两电路各是共射、共集、共基放大电路中的哪一种，并写出 Q 、 A_u 、 R_i 和 R_o 的表达式。

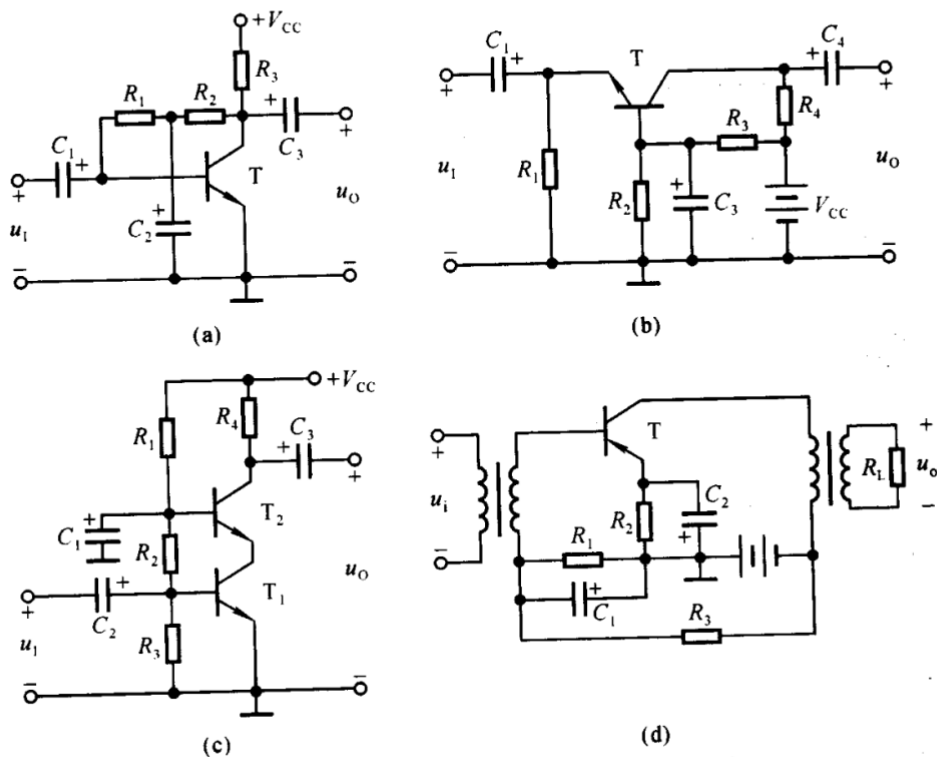


图 P2.2

2.4 Figure P2.4 (a) shows an amplifier circuit. Figure (b) shows the output characteristics of the transistor. The quiescent voltage on BE junction is $U_{BEQ}=0.7V$. Ask: (1) Calculate the quiescent working point and the maximum undistorted output voltage U_{OM} (effective value) at $R_L = \infty$ using the graphic method;

(2) Calculate the quiescent working point and the maximum undistorted output voltage U_{OM} (effective value) at $R_L = 3K \Omega$ using the graphic method.

2.4 电路如图 P2.4 (a)所示, 图(b)是晶体管的输出特性, 静态时 $U_{BEQ} = 0.7 V$ 。利用图解法分别求出 $R_L = \infty$ 和 $R_L = 3 k\Omega$ 时的静态工作点和最大不失真输出电压 U_{om} (有效值)。

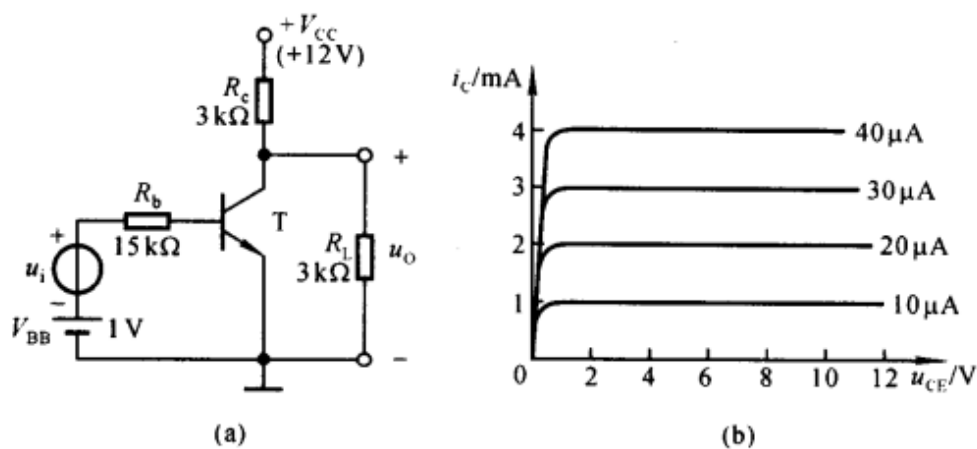


图 P2.4

2.7 An amplifier circuit is shown in figure P2.7. Assume $\beta=80$, $r_{bb'} = 100\Omega$, Ask:

- (1) When $R_L=\infty$, find the quiescent working point (I_{BQ} , I_{CQ} , U_{CEQ}), AC voltage gain A_u , input resistance R_i , and output resistance R_o ;
- (2) When $R_L=3k\Omega$, find the quiescent working point (I_{BQ} , I_{CQ} , U_{CEQ}), AC voltage gain A_u , input resistance R_i , and output resistance R_o .

2.7 电路如图 P2.7 所示，晶体管的 $\beta = 80$, $r_{bb'} = 100 \Omega$ 。分别计算 $R_L = \infty$ 和 $R_L = 3 \text{ k}\Omega$ 时的 Q 点、 A_u 、 R_i 和 R_o 。

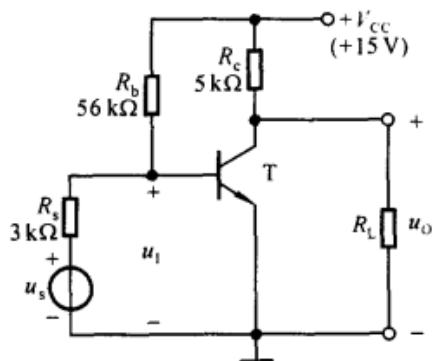


图 P2.7

2.9 For the amplifier circuit shown in figure p2.9, $\beta = 80$ and $r_{be} = 1.4 \text{ k}\Omega$

- (1) What's the values of R_b when $U_{CEQ} = 6 \text{ V}$
- (2) When $u_i = 1 \text{ mV}$, $u_o = 100 \text{ mV}$, what's the value of $R_L = ?$

2.9 已知图 P2.9 所示电路中晶体管的 $\beta = 100$, $r_{be} = 1.4 \text{ k}\Omega$ 。

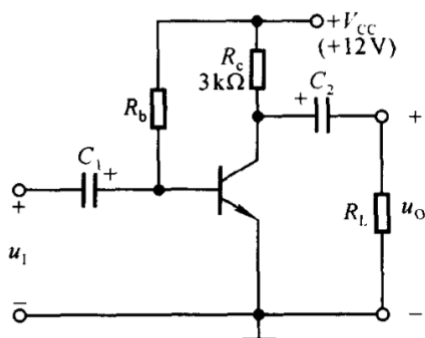


图 P2.9

- (1) 现已测得静态管压降 $U_{CEQ} = 6 \text{ V}$, 估算 R_b 约为多少千欧?
- (2) 若测得 $\dot{U}_i$ 和 $\dot{U}_o$ 的有效值分别为 1 mV 和 100 mV , 则负载电阻 R_L 约为多少千欧?

2.10 The amplifier circuit is shown in Figure P2.9. At quiescent working points $I_{CQ} = 2 \text{ mA}$ and the voltage drop on CE junction when enters into saturation region is $U_{CES} = 0.6 \text{ V}$. Ask:

- (1) When the load resistance $R_L = \infty$, what is the maximum undistorted output voltage of the circuit (dynamic range)?
- (2) When the load resistance $R_L = 3 \text{ k}\Omega$, what is the maximum undistorted output voltage of the circuit (dynamic range)?

2.10 在图 P2.9 所示电路中, 设静态时 $I_{CQ} = 2 \text{ mA}$, 晶体管饱和管压降 $U_{CES} = 0.6 \text{ V}$ 。试问: 当负载电阻 $R_L = \infty$ 和 $R_L = 3 \text{ k}\Omega$ 时电路的最大不失真输出电压各为多少伏?

2.11 The amplifier circuit is shown in figure p2.11. The transistor $\beta=100$, $r_{bb'}=100\Omega$.

(1) Find quiescent working point Q (I_{BQ} , I_{CQ} , U_{CEQ}), AC voltage gain A_u , input resistance R_i , and output resistance R_o ;

(2) If the transistor is replaced with another one with $\beta = 200$, how does the Q point change?

(3) If we remove capacitor C_e , how will dynamic parameters (A_u , R_i and R_o) change?

2.11 电路如图 P2.11 所示, 晶体管的 $\beta = 100$, $r_{bb'} = 100 \Omega$ 。

(1) 求电路的 Q 点、 A_u 、 R_i 和 R_o ;

(2) 若改用 $\beta = 200$ 的晶体管, 则 Q 点如何变化?

(3) 若电容 C_e 开路, 则将引起电路的哪些动态参数发生变化? 如何变化?

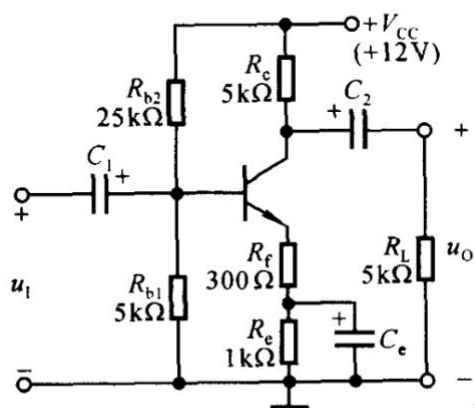


图 P2.11